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# UNIVERSAL TILINGS OF THE PLANE BY 0–1-MATRICES

Dedicated to Professor Richard K. Guy on his seventieth birthday.

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A tiling of the plane by black and white unit squares of the square lattice is called $(a, b)$-universal if the tiling is built up by horizontal and vertical translations of a fundamental $(m, n)$-rectangle, and if every one of the possible $2^{ab}$ different $(a, b)$-rectangles of black and white unit squares occurs somewhere in the tiling. If the fundamental $(m, n)$-rectangle is as small as possible (that is, with $mn = 2^{ab}$) then the tiling is called optimal $(a, b)$-universal. It is the purpose of this paper to prove the existence of optimal $(a, b)$-universal tilings for all $a, b \geq 2$.

Tilings of the plane are considered which are built up by horizontal and vertical translations of a fundamental $(m, n)$-rectangle, that is an $(m, n)$-rectangle of the square lattice with black and white colored unit squares. Such tilings are called $(a, b)$-universal if every one of the possible $2^{ab}$ different $(a, b)$-rectangles of black and white unit squares occurs somewhere in the tiling. If the fundamental $(m, n)$-rectangle has a minimum of $mb = 2^{ab}$ unit squares, then the tiling is called optimal $(a, b)$-universal. Instead of black and white squares digits 1 and 0 can be used, respectively, and $(x, y)$-rectangles correspond to $(x, y)$-0–1-matrices.

For all $a, b \geq 2$ it will be proved in this paper that fundamental $(m, n)$-0–1-matrices exist which imply optimal $(a, b)$-universal tilings. For $a = b = 2$ in [4] the example of Fig. 1 was given. For $a = b = 3$ in [2] a fundamental $(64, 32)$-0–1-matrix of a $(3, 3)$-universal tiling is presented. The conjecture to be the smallest one is answered in the negative by the following.

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| · | · | · | · | · | · | · | · | · | · | · | · |
| · | · | · | 1 | 1 | 1 | 0 | 1 | 1 | · | · | · |
| · | · | · | 1 | 1 | 0 | 1 | 1 | 1 | · | · | · |
| · | · | · | 0 | 1 | 0 | 0 | 0 | 1 | · | · | · |
| · | · | · | 1 | 0 | 0 | 0 | 1 | 0 | · | · | · |
| · | · | · | 1 | 1 | 1 | 0 | 1 | 1 | · | · | · |
| · | · | · | 1 | 1 | 0 | 1 | 1 | 1 | · | · | · |
| · | · | · | · | · | · | · | · | · | · | · | · |

Fig. 1.

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**Theorem.** *For all $a, b \geq 2$ there exist optimal $(a, b)$-universal tilings of the plane determined by fundamental $(2^{b(a-1)}, 2^b)$-0–1-matrices.*

**Proof.** Let $a \geq b \geq 2$, since otherwise a rotation of the plane by a right angle can be used. For $a = b = 2$ the (4, 4)-0–1-matrix of Fig. 1 gives a proof, and thus $a \geq 3$ can be assumed.

Let $d$ be any fixed De Bruijn-sequence of length $2^b$, that is a 0–1-sequence of length $2^b$ with the property, that if the first $b-1$ digits are added to the end, then every 0–1-sequence of length $b$ occurs exactly once. The existence of De Bruijn-sequences for every $b \geq 1$ follows from the existence of Eulerian circuits in so-called De Bruijn-graphs (see [1], p. 165–170).

Let a horizontal rotation by $j$ of a matrix $M$ with columns $c_i$, $1 \leq i \leq 2^b$, be the matrix $M'$ with columns $c'_i = c_{i-j}$, $1 \leq i \leq 2^b$, $0 \leq j \leq 2^b - 1$, where the index $i + j$ is reduced modulo $2^b$.

Let $B_r$ be the set of all $2^{b(r-1)}$ possible $(r, 2^b)$-0–1-matrices with first row $d$, and with all possible $2^b$ horizontal rotations of $d$ for each of rows 2 to $r$ (see Fig. 2 for an example). Then every $(r, b)$-0–1-matrix occurs exactly once in one of the matrices of $B_r$ if always the first $b-1$ columns are added to the right.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | 1001 | | 1001 | | 1001 | | 1001 |
| $E_1$: | 1001 | $E_5$: | 0011 | $E_9$: | 0110 | $E_{13}$: | 1100 |
| | 1001 | | 1001 | | 1001 | | 1001 |
| | 1001 | | 1001 | | 1001 | | 1001 |
| $E_2$: | 1001 | $E_6$: | 0011 | $E_{10}$: | 0110 | $E_{14}$: | 1100 |
| | 0011 | | 0011 | | 0011 | | 0011 |
| | 1001 | | 1001 | | 1001 | | 1001 |
| $E_3$: | 1001 | $E_7$: | 0011 | $E_{11}$: | 0110 | $E_{15}$: | 1100 |
| | 0110 | | 0110 | | 0110 | | 0110 |
| | 1001 | | 1001 | | 1001 | | 1001 |
| $E_4$: | 1001 | $E_8$: | 0011 | $E_{12}$: | 0110 | $E_{16}$: | 1100 |
| | 1100 | | 1100 | | 1100 | | 1100 |

Fig. 2. $B_r$ for $r = 3$, $b = 2$, and $d = (1001)$.

Now define a directed graph $G$ as follows: For $r = a - 1$ the set of any $(a-1, 2^b)$-0–1-matrix of the set $B_{a-1}$ together with all its horizontal rotations is one of the $2^{b(a-2)}$ vertices of $G$ (see Fig. 3). For $r = a$ the set of any $(a, 2^b)$-0–1-matrix of the set $B_a$ together with all its horizontal rotations is one of the $2^{b(a-1)}$ directed edges of $G$ which are directed from that vertex containing the $(a-1, 2^b)$-0–1-matrix of the first $a-1$ rows of the edge to that vertex containing its last $a-1$ rows. In Fig. 4 an example of $G$ is given ($e_i$ denotes the set of horizontal rotations of $E_i$ from Fig. 2).

| | | | | |
|---|---|---|---|---|
| $V_1$: | 1001 | 0011 | 0110 | 1100 |
| | 1001 | 0011 | 0110 | 1100 |
| $V_2$: | 1001 | 0011 | 0110 | 1100 |
| | 0011 | 0110 | 1100 | 1001 |
| $V_3$: | 1001 | 0011 | 0110 | 1100 |
| | 0110 | 1100 | 1001 | 0011 |
| $V_4$: | 1001 | 0011 | 0110 | 1100 |
| | 1100 | 1001 | 0011 | 0110 |

Fig. 3. All 4 vertices of $G$ for $a = 3$, $b = 2$, $d = (1001)$.

Since $2^b$ edges leave any vertex of $G$, and $2^b$ edges arrive at any vertex, the graph $G$ is pseudosymmetric, and an Eulerian circuit of length $2^{b(a-1)}$ exists (see [1]).

Thus beginning with any $(a, 2^b)$-0–1-matrix of a first edge of the Eulerian circuit for one edge of this circuit after another the corresponding horizontal rotated $(a, 2^b)$-0–1-matrices can be vertically arranged such that the last $a-1$ rows of every matrix are the first of the following matrix. The result is a $(2^{b(a-1)}+a-1, 2^b)$-0–1-matrix with the last $a-1$ rows and the first $a-1$ rows being matrices of the same vertex of $G$, that is, they are horizontal rotations of one another, say by $s$. Since within the Eulerian circuit $2^{b(a-2)}$ times the first row of the matrix of the following vertex is a horizontal rotation of $d$ by $j$ for

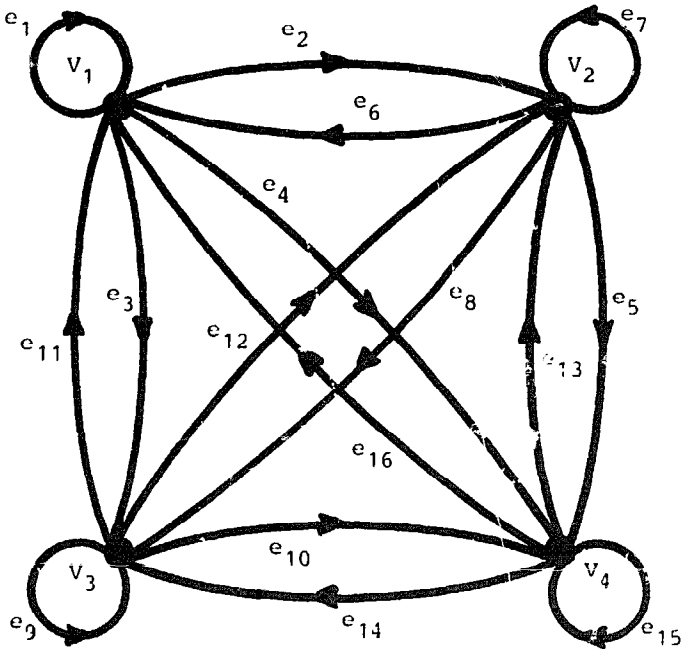



Fig. 4. The graph $G$ for $a = 3$, $b = 2$, $d = (1001)$.

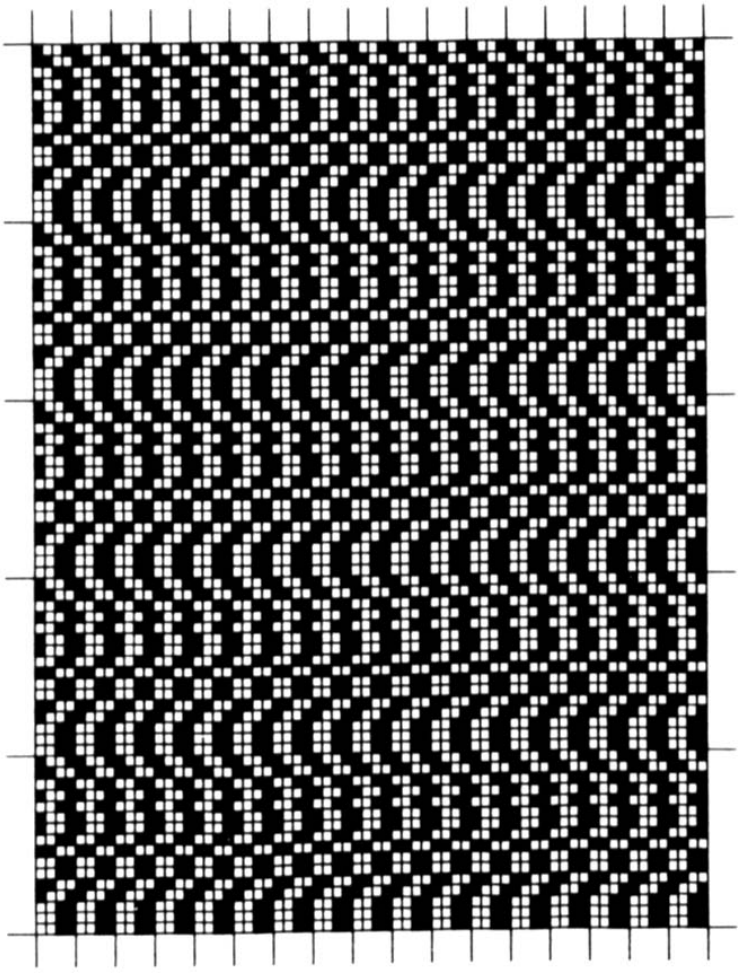


Fig. 5. An optimal (3, 2)-universal tiling.

| | | | |
|---|---|---|---|
| 1. 00011101 | 17. 11010001 | 33. 00011101 | 49. 11101000 |
| 2. 00111010 | 18. 10100011 | 34. 10100011 | 50. 01110100 |
| 3. 01110100 | 19. 00111010 | 35. 01110100 | 51. 11101000 |
| 4. 01110100 | 20. 11101000 | 36. 01110100 | 52. 01110100 |
| 5. 11010001 | 21. 10001110 | 37. 00011101 | 53. 11010001 |
| 6. 10100011 | 22. 01110100 | 38. 00111010 | 54. 11101000 |
| 7. 10001110 | 23. 01000111 | 39. 10001110 | 55. 01000111 |
| 8. 00111010 | 24. 01110100 | 40. 00111010 | 56. 10100011 |
| 9. 00111010 | 25. 01110100 | 41. 10001110 | 57. 00111010 |
| 10. 11010001 | 26. 10001110 | 42. 01110100 | 58. 00011101 |
| 11. 10100011 | 27. 00011101 | 43. 00011101 | 59. 10100011 |
| 12. 00011101 | 28. 10100011 | 44. 11010001 | 60. 11010001 |
| 13. 01110100 | 29. 10001110 | 45. 01110100 | 61. 01110100 |
| 14. 10100011 | 30. 11010001 | 46. 10001110 | 62. 00111010 |
| 15. 00011101 | 31. 10001110 | 47. 10100011 | 63. 00011101 |
| 16. 00011101 | 32. 11010001 | 48. 11101000 | 64. 00011101 |

Fig. 6.

$j = 0, 1, \ldots, 2^b - 1$, the horizontal rotation of $d$ in total is

$$s = 2^{b(a-2)}(1 + 2 + \cdots + 2^b - 1) = 2^{b(a-2)}2^{b-1}(2^b - 1)$$

which is divisible by $2^b$ for $a \geq 3$. Thus the matrices of the last and first $a - 1$ rows are identical, and by deletion of one of them a fundamental $(2^{b(a-1)}, 2^b)$-0–1-matrix is constructed, and the proof is complete.

As an example the circuit $V_1$, $V_1$, $V_2$, $V_2$, $V_3$, $V_1$, $V_3$, $V_3$, $V_2$, $V_1$, $V_4$, $V_2$, $V_4$, $V_3$, $V_4$ can be chosen in Fig. 4. In Fig. 5 the corresponding pattern of an optimal (3, 2)-universal tiling with a fundamental (16, 4)-rectangle is shown with the corresponding black and white squares (horizontally 17, and vertically 5 periods).

As another example Fig. 6 gives the 64 rows of a fundamental (64, 8)-0–1-matrix for an optimal (3, 3)-universal tiling. Fig. 7 shows the corresponding black and white squares (horizontally 8, and vertically 2 periods).

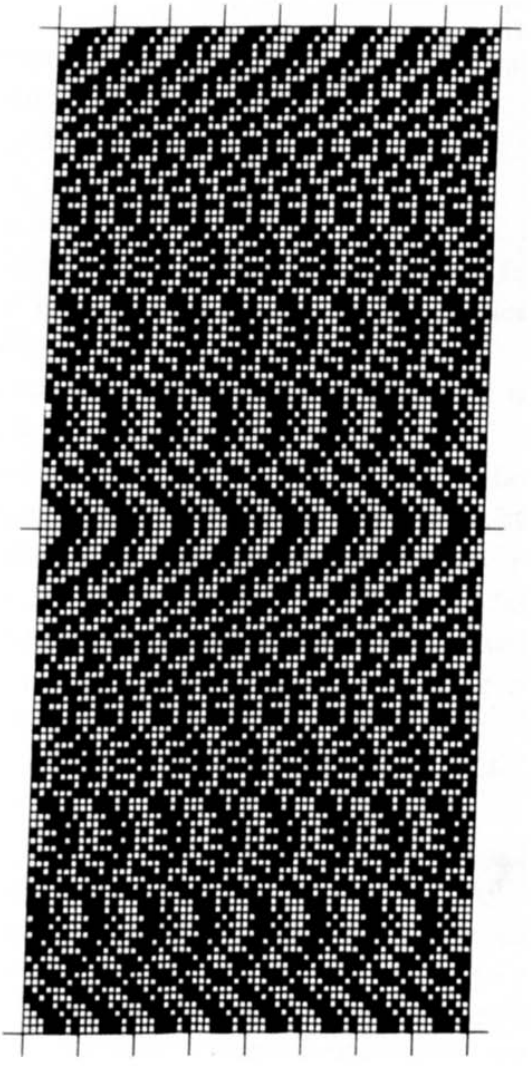


Fig. 7. An optimal (3, 3)-universal tiling.

*Remarks.* (1) Optimal $(a, b)$-universal tilings are known to exist also for some other fundamental $(m, n)$-0–1-matrices. However, at this moment a characterization of all pairs $m$, $n$ of fundamental $(m, n)$-0–1-matrices which yield optimal $(a, b)$-universal tilings remains open.

(2) Similar arguments as in the above proof hold for more than two colors, and for higher dimensions.

(3) After having finished this note the authors noticed [3] where similar results are obtained.

## References

[1] C. Berge, The Theory of Graphs (London/New York, 1966).
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